

SEC P vs NP log 2026-05-10

SEC (autonomous) — attributed to Ludovico Kubler

2026-05-10 04:00 UTC

SEC P vs NP — daily log 2026-05-10

10 cycles recorded

Tropical Convex Hull Extreme Points Lower Bound for ACC^0 Circuit Size

- **Verdict:** INCONCLUSIVE
- **Bridge:** Tropical Convex Geometry $\times$ ACC^0 Circuit Size
- **Recorded:** 2026-05-10 00:08 UTC
- **Entry ID:** 0bb43dada997

Statement

For any CNF formula Φ with n variables, the number of extreme points in the tropical convex hull of its clause-indicator polynomial is $\Omega(\log n)$ if Φ has no ACC^0 circuit of size n^c for any constant $c > 0$.

Rationale

Tropical convex hulls encode clause interactions via min-plus geometry, and their extreme points reflect combinatorial constraints on circuit representations. ACC^0 circuits' limited depth imposes structural bounds on these hulls, creating a bridge between algebraic geometry and circuit complexity.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c44b66b7.py", line 78, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c44b66b7.py", line 56, in run_trial
    poly = clause_indicator_polynomial(clauses, n)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c44b66b7.py", line 42, in clause_indica
    poly = [0] * (2 ** n)
           ~~~^~~~~~
```

MemoryError

Judge reasoning

Test crashed with MemoryError before producing results, preventing evaluation of conjecture support/falsification. | next: Optimize clause_indicator_polynomial to handle larger n via sparse representations or incremental construction

Kronecker Coefficient Exponential Gap in Symmetric Powers of Permanent vs Determinant

- **Verdict:** INCONCLUSIVE
- **Bridge:** Representation Theory of Symmetric Groups × Communication Complexity of Disjointness
- **Recorded:** 2026-05-10 00:22 UTC
- **Entry ID:** 722bbadc85c5

Statement

For all $n \leq 40$, the Kronecker coefficient for the decomposition of $\text{Sym}^k(\text{Sym}_n(V))$ into irreducible representations is exponentially larger for the permanent than for the determinant when $k = \lceil n/2 \rceil$, implying a $\Omega(n)$ communication complexity lower bound for the corresponding disjointness instance.

Rationale

Kronecker coefficients capture the structure of symmetric power decompositions, which are central to GCT's orbit closure separations. Linking them to communication complexity provides a bridge between algebraic geometry and combinatorial lower bounds, leveraging the exponential sensitivity of symmetric tensors to combinatorial constraints.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

```
nterexample': 'Ratio 1.0 is not exponentially larger than 2^7'}
TRIAL: {'metric_name': 'Kronecker Coefficient Ratio', 'metric_value': 1.0, 'instances_tested': 1, 'co
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TRIAL: {'metric_name': 'Kronecker Coefficient Ratio', 'metric_value': 1.0, 'instances_tested': 1, 'co
TRIAL: {'metric_name': 'Kronecker Coefficient Ratio', 'metric_value': 1.0, 'instances_tested': 1, 'co
```

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_4e40f8cb.py", line 81, in <module>
    first_failing_seed = next(r["seed"] for r in res
```

Judge reasoning

Test crashed before completing evaluation, but partial results show counterexamples where ratio 1.0 fails to meet exponential thresholds. | next: Fix seed handling in test code and re-run with full coverage of $n \leq 40$ cases

Operator Norm Lower Bound for Disjointness Communication Matrices

- **Verdict:** INCONCLUSIVE
- **Bridge:** Operator Spaces Theory $\times$ Communication Complexity of DISJOINTNESS
- **Recorded:** 2026-05-10 00:30 UTC
- **Entry ID:** 4da9869ae44e

Statement

For any $n \times n$ matrix M representing the DISJOINTNESS function, the completely bounded norm $\|M\|_{cb} \geq \Omega(n)$. This holds for all matrices where $M[i,j] = 1$ iff the i -th and j -th subsets are disjoint, with random subset generation using uniform distribution over $2^{\lceil \log_2 n \rceil}$ -element sets.

Rationale

Operator spaces theory provides tools to analyze tensor norms that capture the structure of communication matrices. The cb -norm, being invariant under tensor products, could expose inherent dimensionality constraints in DISJOINTNESS that classical discrepancy methods miss.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_dc72256d.py", line 78, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_dc72256d.py", line 60, in run_trial
    cb_norm_max, _ = numerical_range(M)
                     ^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_dc72256d.py", line 46, in numerical_range
    B = [M[i] - (i+1)*I for i in range(n)]
         ~~~~^~~~~~
```

TypeError: unsupported operand type(s) for -: 'list' and 'list'

Judge reasoning

Test crashed due to TypeError in matrix operations, preventing result collection | next: Fix the numerical_range function's matrix subtraction logic to handle list operations properly

Tensor Rank Exponential Gap in Read-Twice vs Read-Once Branching Programs for IP 2

- **Verdict:** INCONCLUSIVE
- **Bridge:** Tensor Rank over Real Numbers $\times$ Read-Twice Branching Programs
- **Recorded:** 2026-05-10 00:53 UTC
- **Entry ID:** ec8a0368610a

Statement

For any read-once branching program P over n variables, the tensor rank of the transition tensor $T(P)$ is $O(\log n)$. For the IP_2 function, any read-twice branching program Q must satisfy $\text{rank}(T(Q)) \geq 2^{\lfloor n/2 \rfloor} / \text{poly}(n)$.

Rationale

Tensor rank captures the algebraic complexity of branching program transitions. Read-once BPs decompose as low-rank tensors, while read-twice BPs require high-rank tensors to model their repeated access to variables, mirroring the exponential gap in communication complexity for IP 2.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

```
--STDERR--
```

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6ec1712e.py", line 135, in <module>
    result = run_trial(seed)
            ^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6ec1712e.py", line 90, in run_trial
    read_once_matrix = matrix_multiplication(*read_once_bp)
                                ~~~~~
```

```
TypeError: matrix multiplication() takes 2 positional arguments but 5 were given
```

Judge reasoning

Test crashed due to incorrect argument count in matrix multiplication, preventing result collection
| next: Fix matrix multiplication argument count in test 6ec1712e.py

Projective Plane Line Count Bounds ABP Width for Monotone Circuits

- **Verdict:** INCONCLUSIVE
- **Bridge:** Finite Geometry $\times$ Monotone Boolean Circuit Size
- **Recorded:** 2026-05-10 01:02 UTC
- **Entry ID:** 96cc863111fe

Statement

For every monotone Boolean circuit computing the characteristic function of a projective plane $\text{PG}(2, q)$, the minimal width of an ABP computing the same function satisfies $\text{width} \geq q^2 + q + 1$. This bound is tight for $q = 2^k$.

Rationale

Projective planes exhibit highly symmetric incidence structures that could encode combinatorial constraints mirroring the layered computation of ABPs. The tightness of the bound suggests a deep algebraic relationship between geometric incidence and algebraic computation.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_55481e38.py", line 112, in <module>
    result = run_trial(seed)
            ^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_55481e38.py", line 80, in run_trial
    points, lines = generate_projective_plane(q)
                    ^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_55481e38.py", line 20, in generate_proj
    raise ValueError("q must be at least 2")
ValueError: q must be at least 2
```

Judge reasoning

Test crashed due to invalid q value (must be ≥ 2), preventing evaluation of conjecture | next: Run test with q=2 and valid projective plane parameters to verify bounds

Quantum Rank Lower Bounds for Disjointness Communication Matrices

- **Verdict:** INCONCLUSIVE
- **Bridge:** Noncommutative Geometry $\times$ Communication Complexity of DISJOINTNESS
- **Recorded:** 2026-05-10 01:08 UTC
- **Entry ID:** 8dbbdbbc746a

Statement

For the $n \times n$ DISJOINTNESS matrix M_n , the noncommutative rank $\tau(M_n)$ satisfies $\tau(M_n) \geq c \cdot n$ for some universal constant $c > 0$. This invariant is defined as the minimal k such that M_n can be expressed as a sum of k rank-1 matrices in the matrix algebra over the noncommutative ring $\mathbb{C}\langle x, y \rangle / (x^2 = x, y^2 = y)$.

Rationale

Noncommutative geometry provides tools to analyze matrix factorizations through ring-theoretic structures. The DISJOINTNESS matrix's inherent combinatorial structure may enforce a nontrivial lower bound on its noncommutative rank, revealing algebraic obstructions to efficient communication protocols.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_82a6913d.py", line 91, in <module>
    result = run_trial(seed)
            ^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_82a6913d.py", line 67, in run_trial
    tau_M_n = noncommutative_rank(M_n, n)
            ^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_82a6913d.py", line 58, in noncommutative_rank
    A = [[x[i][j] - y[j] * M[i][n + j] for j in range(n)] for i in range(n)]
        ^^^^^^^^^^^^^^^^^^^^^^^^^
```

TypeError: unsupported operand type(s) for -: 'int' and 'list'

Judge reasoning

Test crashed due to TypeError, preventing result collection. Pre-registered support condition cannot be evaluated without successful trials. | next: Debug the noncommutative_rank function's matrix construction to resolve type mismatches in the algebraic operations

Symmetric Tensor Rank Gap in Permanent vs Determinant Decompositions

- **Verdict:** INCONCLUSIVE
- **Bridge:** Representation Theory of Symmetric Groups × Monotone Circuit Size for 3-CNFs
- **Recorded:** 2026-05-10 01:23 UTC
- **Entry ID:** 8de14a1366b2

Statement

For a random 3-CNF formula Φ with n variables, the dimension of the irreducible constituent λ in the symmetric square of the permanent's representation (computed via Young diagrams) exceeds the dimension of the corresponding constituent in the determinant's representation by a factor of $\Omega(2^{\{n/4\}})$ with probability ≥ 0.95 over Φ .

Rationale

Symmetric tensor decompositions reveal structural asymmetry between permanent and determinant. The Young diagram dimensions capture combinatorial complexity, and their exponential gap could imply inherent circuit size separation via representation-theoretic obstructions.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6dfe40e5.py", line 110, in <module>
    result = run_trial(seed)
            ^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6dfe40e5.py", line 83, in run_trial
    perm_val = permanent(n, perm_matrix)
    ~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6dfe40e5.py", line 42, in permanent
    submatrix = [[matrix[i][k] for k in range(j) + range(j+1, n)] for i in range(1, n)]
                  ~~~~~^~~~~~
```

TypeError: unsupported operand type(s) for +: 'range' and 'range'

Judge reasoning

Test crashed due to Python TypeError in range concatenation, preventing data collection. Support condition (80% success rate) cannot be evaluated without valid results. | next: Fix the range concatenation error in the permanent function by using list concatenation (e.g., list(range(j)) + list(range(j+1, n)))

Negative Eigenvalue Count Lower Bounds SOS Degree for Max-CUT

- **Verdict:** INCONCLUSIVE
- **Bridge:** Real Algebraic Geometry × SOS Degree for Max-CUT
- **Recorded:** 2026-05-10 01:47 UTC
- **Entry ID:** 510fd15279df

Statement

For any n -variable Max-CUT instance, the minimal SOS degree d required to certify its integrality gap satisfies $d \geq k + 1$, where k is the number of negative eigenvalues of the moment matrix of the degree- d pseudoexpectation.

Rationale

The number of negative eigenvalues quantifies the ‘deficiency’ in positive semidefiniteness of the moment matrix, which directly correlates with the need for higher-degree SOS constraints to resolve non-positivity in the polynomial representation of the Max-CUT objective.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.14s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_0ee3b164.py", line 116, in <module>
    result = run_trial(seed)
    ~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_0ee3b164.py", line 91, in run_trial
    eigenvalues = [abs(e) for e in gaussian_elimination(M)[-1]]
                  ~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_0ee3b164.py", line 28, in gaussian_elim
    factor = 1 / A[i][i]
             ~^~~~~~
```

ZeroDivisionError: division by zero

Judge reasoning

Test crashed with ZeroDivisionError, preventing data collection. Pre-registered support condition cannot be evaluated without valid results. | next: Debug the gaussian_elimination function to handle singular matrices, then re-run tests with multiple seeds

Plethysm Coefficient Gap in Homogeneous Polynomial Decompositions

- **Verdict:** INCONCLUSIVE
- **Bridge:** Plethysm Theory $\times$ Boolean Circuit Size
- **Recorded:** 2026-05-10 03:39 UTC
- **Entry ID:** cc61e4e0da28

Statement

For any $n \geq 2$, the plethysm coefficient of the permanent polynomial perm_n decomposed into symmetric squares of $\text{GL}(n)$ representations exceeds that of det_m for all $m < n^{\{1.5\}}$, with ratio $\geq 2^{\{\Omega(n)\}}$

Rationale

Plethysm coefficients encode symmetry-breaking patterns in polynomial representations. If permanent's decomposition requires exponentially more symmetric squares than determinant's, this could expose structural asymmetries exploitable for circuit lower bounds.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 124, elapsed: 240.00s

TIMEOUT after 240s

Judge reasoning

Test timed out before producing results, preventing verification of support fraction or counterexamples. | next: Run test with extended timeout and higher resource allocation

Hypergraph Maximum Matching Inverse Proportional to ACC^0 Circuit Size for 3-CNFs

- **Verdict:** INCONCLUSIVE
- **Bridge:** Hypergraph Theory $\times$ ACC^0 Circuit Size
- **Recorded:** 2026-05-10 03:58 UTC
- **Entry ID:** 9d4ad01564fe

Statement

For any 3-CNF formula with n variables, the maximum matching size of its clause hypergraph is $\Theta(\log n)$ if and only if the formula can be computed by an ACC^0 circuit of size $\Theta(n^k)$ for some constant k .

Rationale

The maximum matching size captures the sparsity of clause overlaps. Sparse overlaps (logarithmic) suggest a structured dependency that can be captured by ACC⁰ circuits, which have limited depth but can handle structured patterns.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.10s

```

'max_matching_size', 'metric_value': 85, 'instances_tested': 1, 'conjecture_holds': False, 'countere
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```

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_70a9fc8c.py", line 82, in <module>
    first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
                        ~~~~~^~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_70a9fc8c.py", line 82, in <genexpr>
    first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
                        ~~~~~^~~~~~
```

KeyError: 'seed'

Judge reasoning

Test crashed before completing validation; counterexamples in output may be unreliable | next:
Re-run test with increased time limits and deterministic seed configuration